EMATH 100

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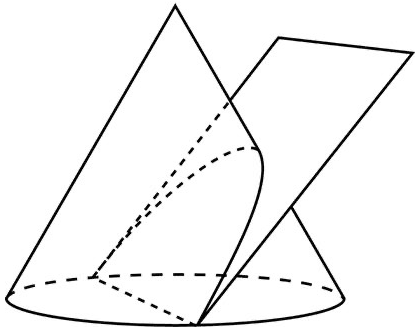
PARABOLAS

Graphing Quadratic Functions

$$y = ax^2 + bx + c$$

Parabola

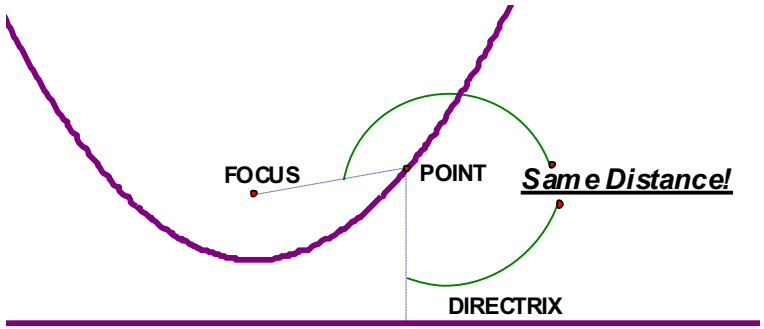
- **Review: The geometric definition relies on a cone and a plane intersecting it.**



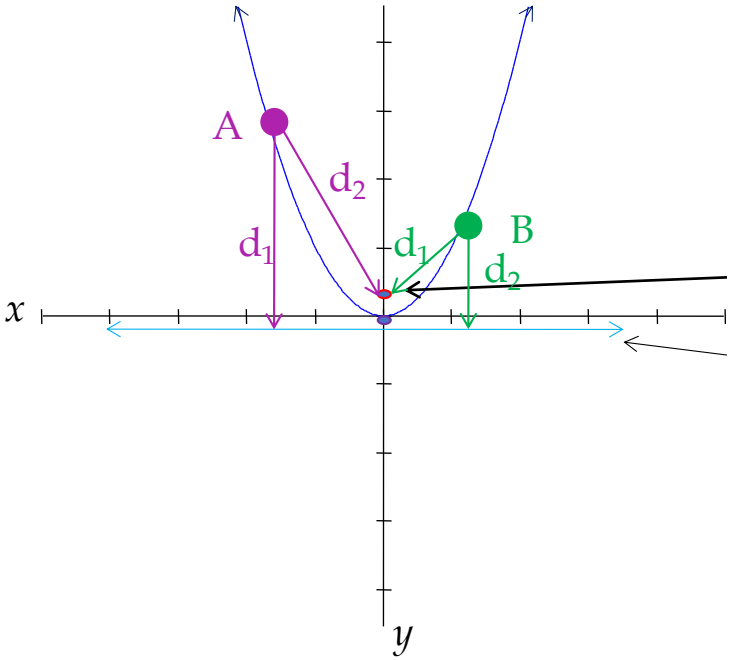
- **Algebraic definition: All points that are equidistant from a given line (the directrix)and a fixed point not on the directrix (the focus)**

Definition

- The set of all points in a plane that are the same distance from a given point called the focus and a given line called the directrix.



Parabola



Focus

Directrix

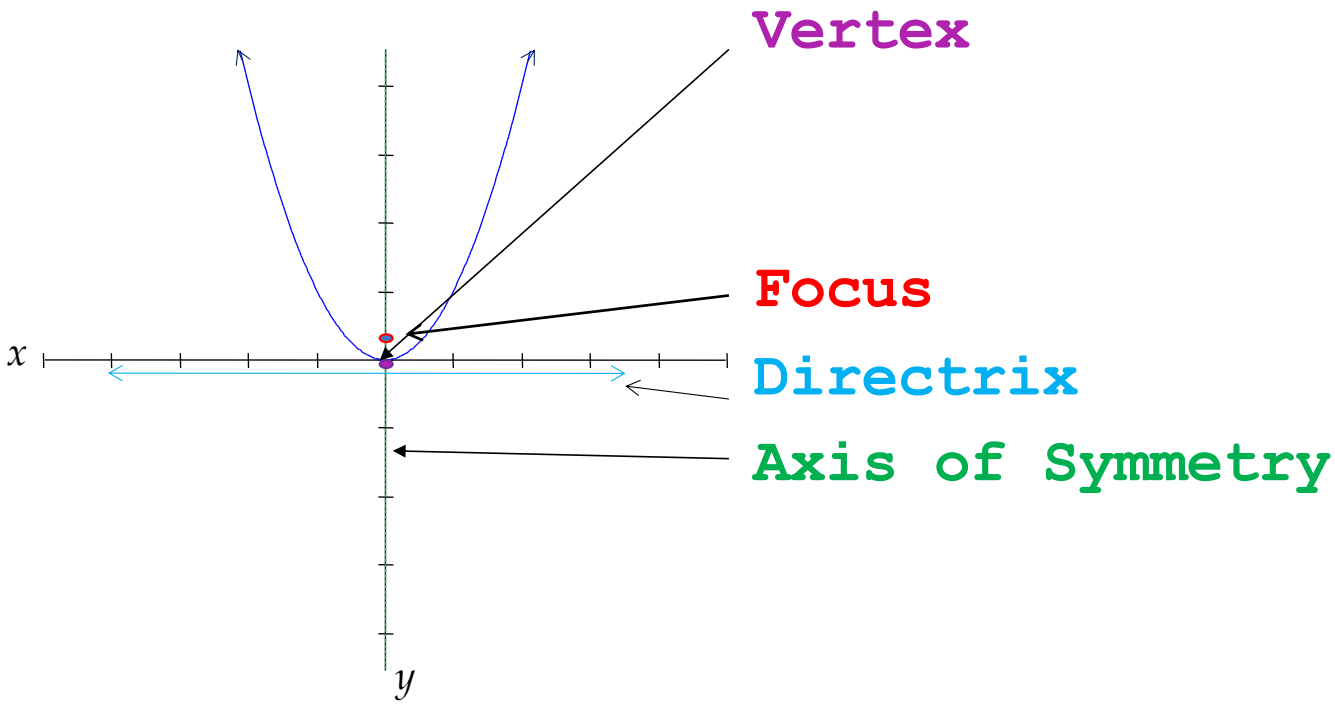
Any point on the parabola is equidistant to the focus and the directrix.

Example:

Point A: $d_1=d_2$

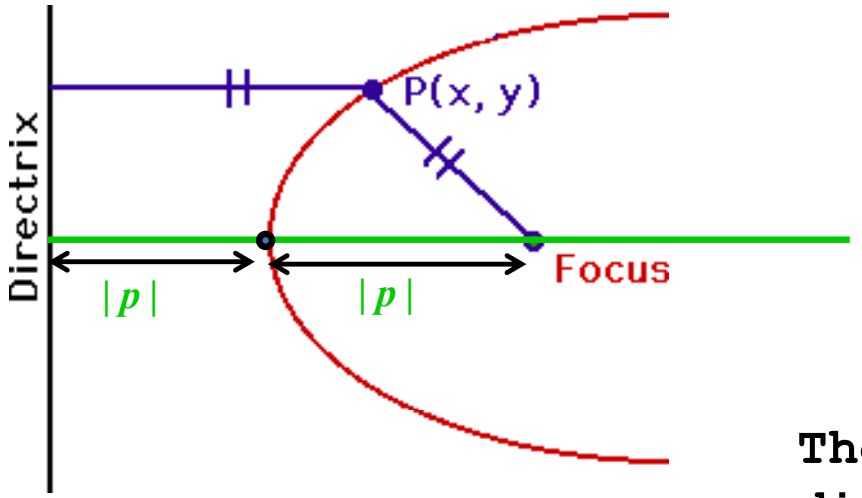
Point B: $d_1=d_2$

Items referenced on the graph of a parabola:



The Parabola

The parabola is the locus of all points in a plane that are the same distance from a line in the plane, the **directrix**, as from a fixed point in the plane, the **focus**.



Point Focus = Point Directrix

PF = PD

The parabola has one **axis of symmetry**, which intersects the parabola at its **vertex**.

The distance from the vertex to the focus is **| p |**.

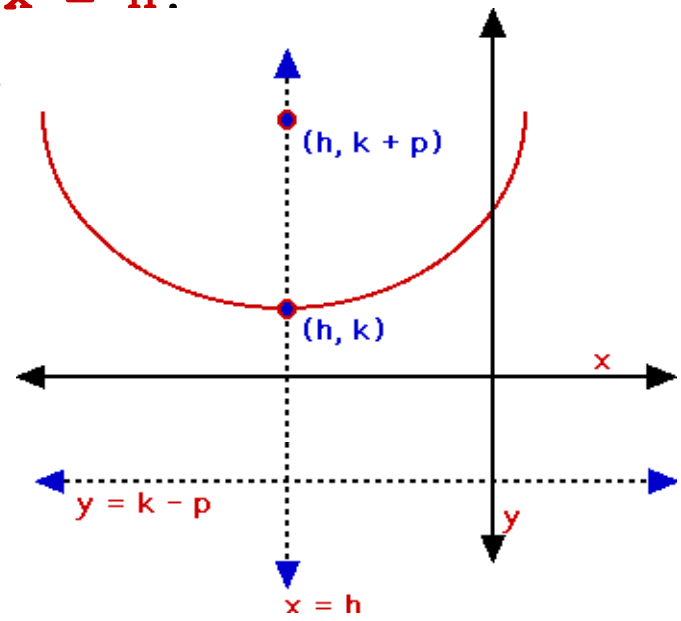
The distance from the directrix to the vertex is also **| p |**.

The Standard Form of the Equation with Vertex (h, k)

For a parabola with the axis of symmetry parallel to the y-axis and vertex at (h, k), the standard form is …

$$(x - h)^2 = 4p(y - k)$$

- The equation of the axis of symmetry is x = h.
- The coordinates of the focus are (h, k +
- The equation of the directrix is y = k - p.
- When p is positive, the parabola opens upward.
- When p is negative, the parabola opens downward.

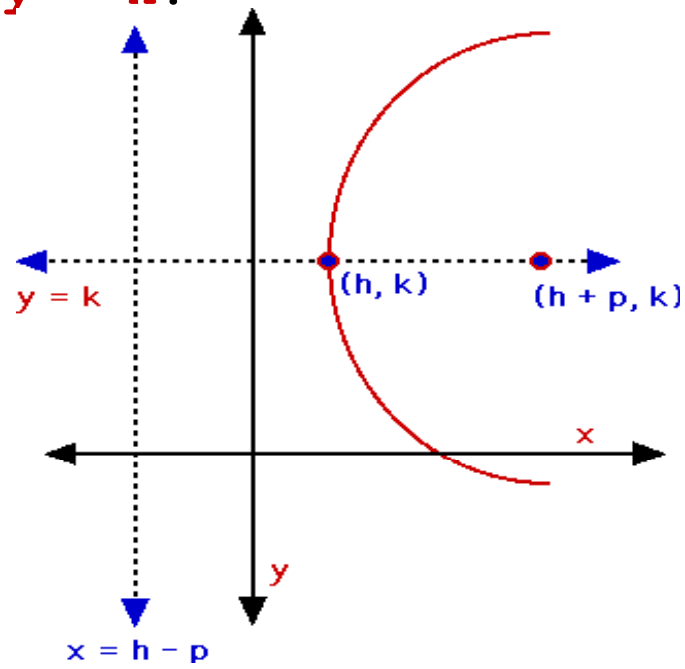


The Standard Form of the Equation with Vertex (h, k)

For a parabola with an axis of symmetry parallel to the x-axis and a vertex at (h, k), the standard form is:

$$(y - k)^2 = 4p(x - h)$$

- The equation of the axis of symmetry is $y = k$.
- The coordinates of the focus are $(h + p, k)$.
- The equation of the directrix is $x = h - p$.
- When p is positive, the parabola opens to the right.
- When p is negative, the parabola opens to the left.

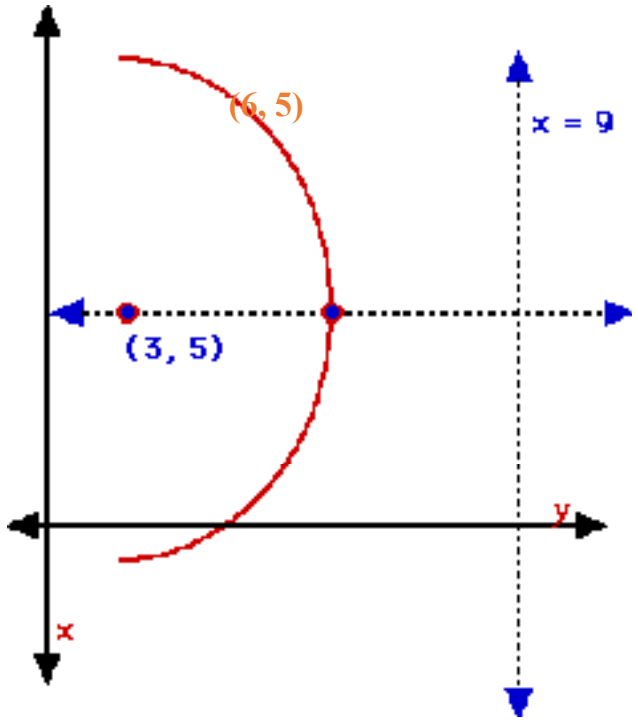


Finding the Equations of Parabolas

Write the equation of the parabola with a focus at (3, 5) and the directrix at x = 9, in standard form and general form

The distance from the focus to the directrix is 6 units, therefore, 2p = -6, p = -3.

Thus, the vertex is (6, 5).



The axis of symmetry is parallel to the x-axis:

$(y - k)^2 = 4p(x - h)$

$(y - 5)^2 = 4(-3)(x - 6)$ h = 6 and k = 5

$(y - 5)^2 = -12(x - 6)$ Standard form

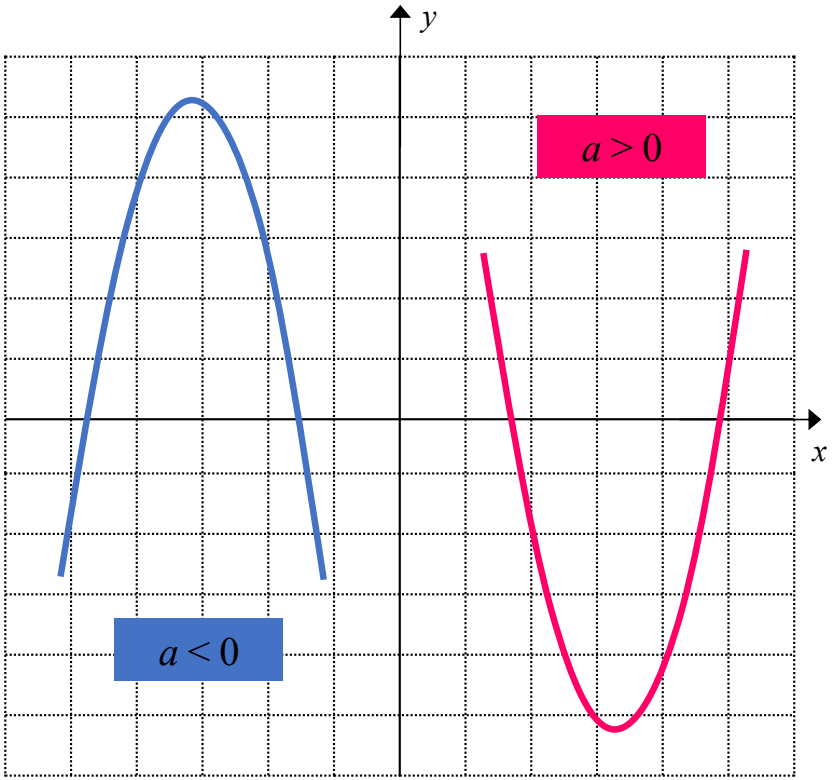
Standard Form

The standard form of a quadratic function is

$$y = ax^2 + bx + c$$

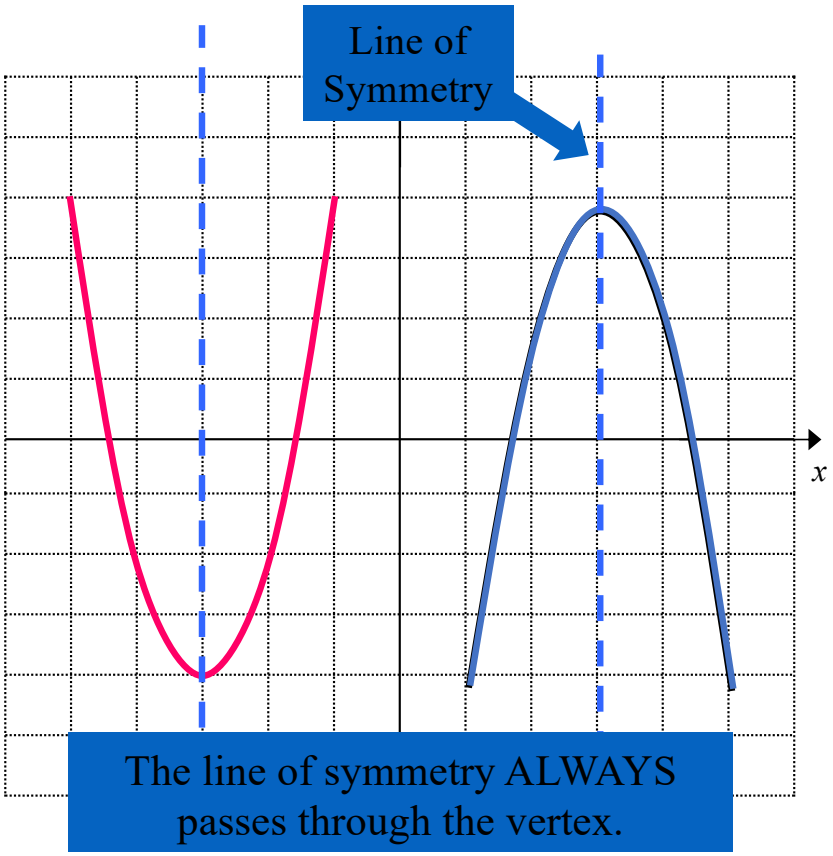
The parabola will open up when the a value is positive.

The parabola will open down when the a value is negative.



Line of Symmetry

- Parabolas have a symmetric property to them.
- If we drew a line down the middle of the parabola, we could fold the parabola in half.
- We call this line the line of symmetry.
- Or, if we graphed one side of the parabola, we could "fold" (or REFLECT) it over, the line of symmetry to graph the other side.



Finding the Line of Symmetry

When a quadratic function is in standard form

$y = ax^2 + bx + c$,

The equation of the line of symmetry is

$$x = \frac{-b}{2a}$$

This is best read as …

the opposite of b divided by the quantity of 2 times a.

For example…

Find the line of symmetry of

$y = 3x^2 - 18x + 7$

Using the formula…

$$x = \frac{18}{2(3)} = \frac{18}{6} = 3$$

Thus, the line of symmetry is x = 3.

Finding the Vertex

We know the line of symmetry always goes through the vertex.

Thus, the line of symmetry gives us the x – coordinate of the vertex.

To find the y – coordinate of the vertex, we need to plug the x – value into the original equation.

$$y = -2x^2 + 8x - 3$$

STEP 1: Find the line of symmetry

$$x = \frac{-b}{2a} = \frac{-8}{2(-2)} = \frac{-8}{-4} = 2$$

STEP 2: Plug the x – value into the original equation to find the y value.

$$y = -2(2)^2 + 8(2) - 3$$

$$y = -2(4) + 8(2) - 3$$

$$y = -8 + 16 - 3$$

$$y = 5$$

Therefore, the vertex is (2 , 5)

Writing Linear Equation in Parabolic Form

- **GOAL: Turn** $y = ax^2 + bx + c$ into

$$y = a(x - h)^2 + k$$

Writing Linear Equation in Parabolic Form

1. Start with $y = ax^2 + bx + c$
2. Group the two x-terms
3. Pull out the constant with x^2 from the grouping
4. Complete the square of the grouping
5. Write the squared term as subtraction so that you end with $y = a(x - h)^2 + k$

$$Example: y = 3x^2 + 12x + 16$$

$y = (3x^2 + 12x) + 16$ **Group x-terms**

$y = 3(x^2 + 4x) + 16$ **Pull out GCF**

$y = 3(x^2 + 4x + ___) + 16 - ___$ **Complete the Square**

Remember that whatever you add in the grouping must be subtracted from the c-value

$y = 3(x^2 + 4x + 4) + 16 - 12$

$y = 3(x + 2)^2 + 4$ or **Factor and simplify**

$y = 3(x - (-2))^2 + 4$

A Quadratic Function in Standard Form

The standard form of a quadratic function is given by

$$y = ax^2 + bx + c$$

STEP 1: Find the line of symmetry

STEP 2: Find the vertex

STEP 3: Find two other points and reflect them across the line of symmetry. Then connect the five points with a smooth curve.

There are 3 steps to graphing a parabola in standard form.

MAKE A TABLE

using x – values close to the line of symmetry.

Plug in the line of symmetry (x – value) to obtain the y – value of the vertex.

USE the equation

$$x = \frac{-b}{2a}$$

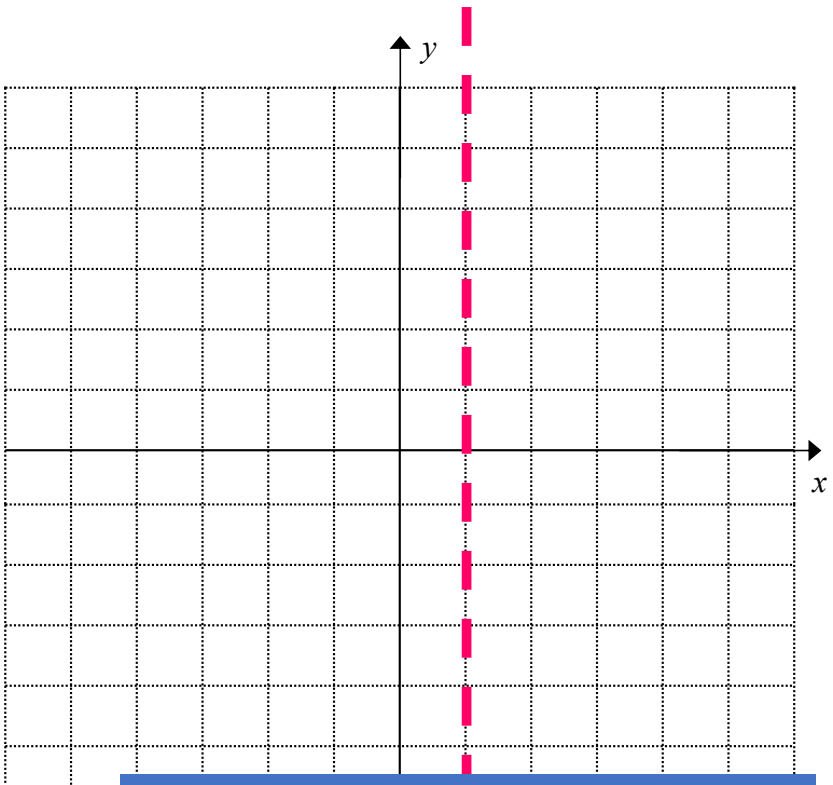
A Quadratic Function in Standard Form

Let's Graph ONE!
Try …

$$y = 2x^2 - 4x - 1$$

STEP 1: Find the line of symmetry

$$x = \frac{-b}{2a} = \frac{4}{2(2)} = 1$$



Thus the line of symmetry is x = 1

A Quadratic Function in Standard Form

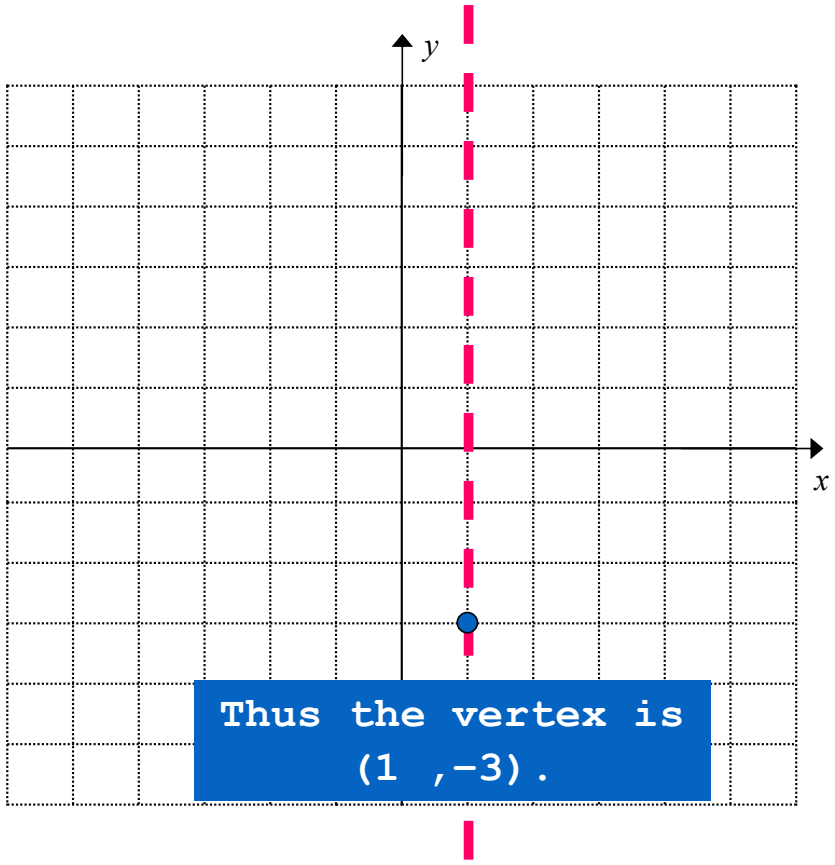
Let's Graph ONE!
Try …

$$y = 2x^2 - 4x - 1$$

STEP 2: Find the vertex

Since the x – value of the vertex is given by the line of symmetry, we need to plug in x = 1 to find the y – value of the vertex.

$$y = 2(1)^2 - 4(1) - 1 = -3$$



A Quadratic Function in Standard Form

Let's Graph ONE!
Try …

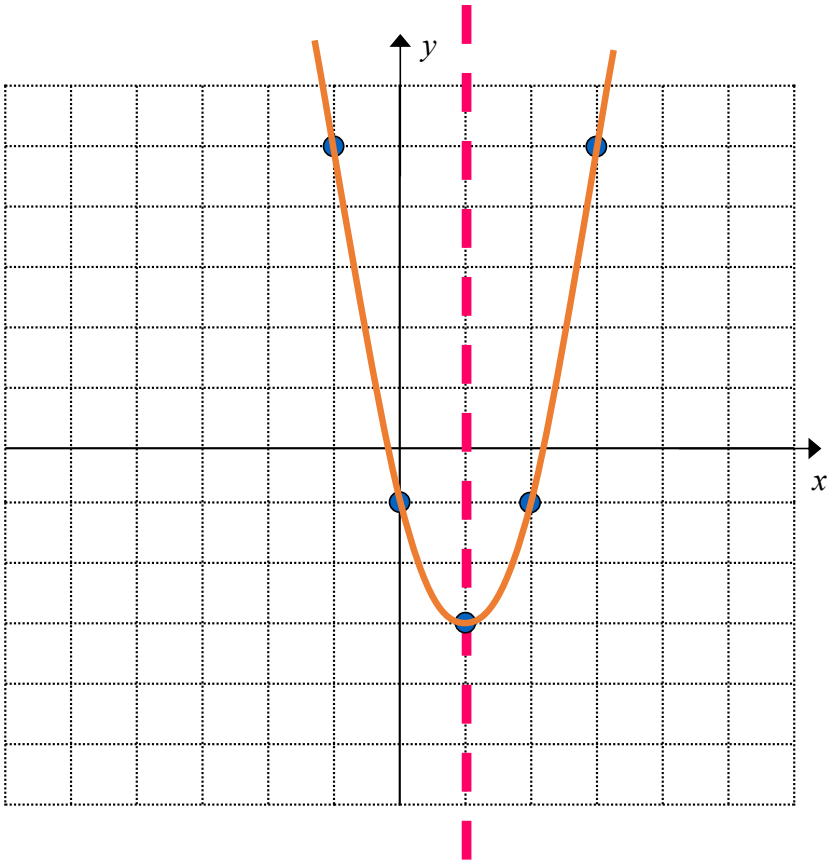
$$y = 2x^2 - 4x - 1$$

STEP 3: Find two other points and reflect them across the line of symmetry. Then connect the five points with a smooth curve.

$y = 2(2)^2 - 4(2) - 1 = -1$

$y = 2(3)^2 - 4(3) - 1 = 5$

x	y
2	–1
3	5



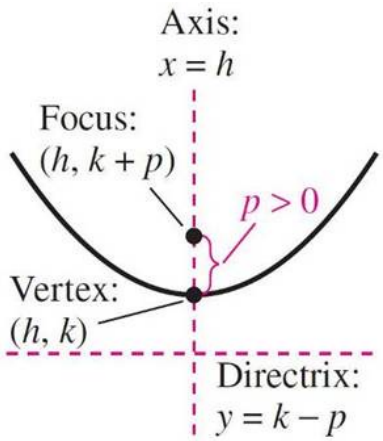
Why write in parabolic form?

v

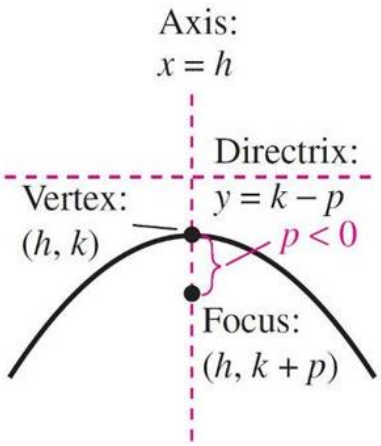
It gives you necessary information to draw the parabola

Equation	$y = a(x-h)^2 + k$	$x = a(y-k)^2 + h$
Axis of symmetry	$x = h$	$y = k$
Vertex	(h, k)	(h, k)
Focus	$\left(h, k + \frac{1}{4a}\right)$	$\left(h + \frac{1}{4a}, k\right)$
Directrix	$y = k - \frac{1}{4a}$	$x = h - \frac{1}{4a}$
Direction of opening	Up: a>0, Down: a<0	Right: a>0, Left: a<0
Latus Rectum	$\left\|\frac{1}{a}\right\|$ units	$\left\|\frac{1}{a}\right\|$ units

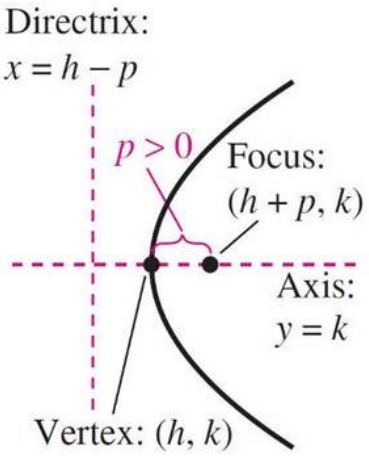
Parabolas



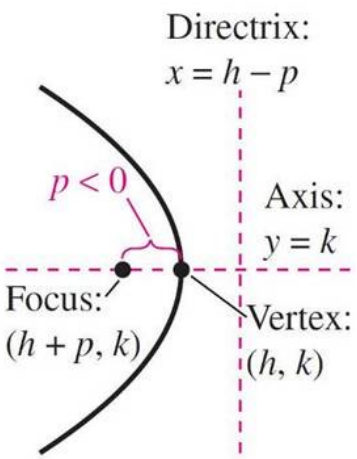
$(x - h)^2 = 4p(y - k)$



$(x - h)^2 = 4p(y - k)$



$(y - k)^2 = 4p(x - h)$

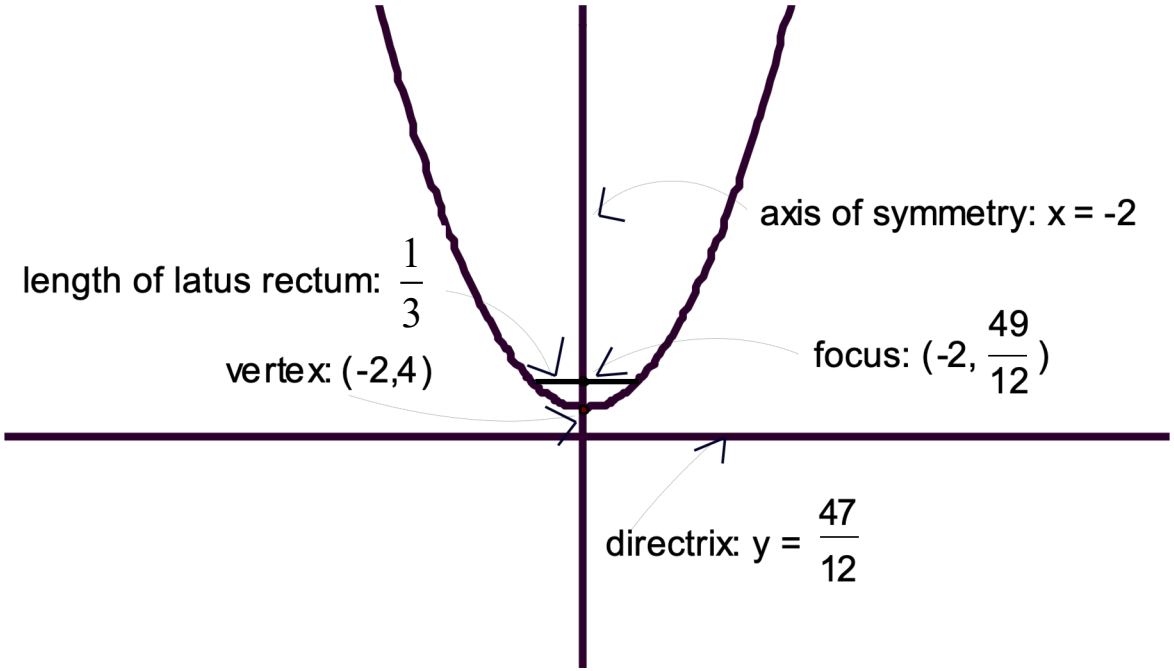


$(y - k)^2 = 4p(x - h)$

(a) Vertical axis: p > 0 **(b) Vertical axis: p < 0** **(c) Horizontal axis: p > 0** **(d) Horizontal axis: p < 0**

Graph of prior example

$$y = 3(x-(-2))^2 + 4$$



You Try!!

Write the following equation in parabolic form. State the vertex, axis of symmetry and direction of opening.

$$x = y^2 - 10y - 7$$

A QUADRATIC FUNCTION

- A quadratic function has a form

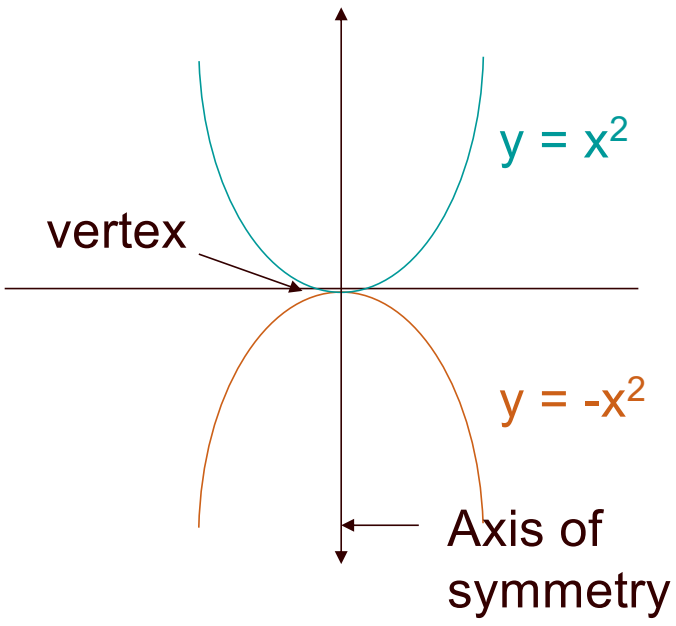
  $y = ax^2 + bx + c$

  where $a \neq 0$.

- The graph of a quadratic function is U-shaped and is called parabola.

- The lowest or highest point on the graph of a quadratic function is called the vertex.

- The graphs of $y = x^2$ and $y = -x^2$ are symmetric about the y-axis, called the axis of symmetry.



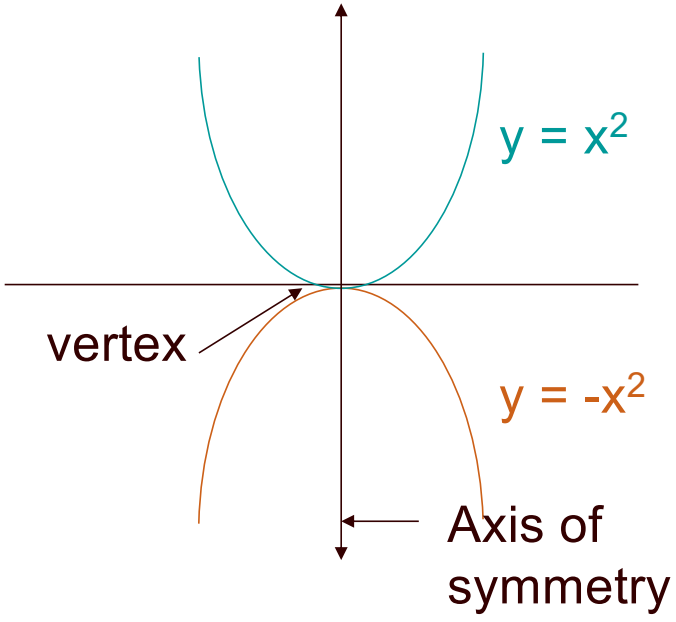
THE GRAPH OF A QUADRATIC FUNCTION

- The parabola opens up if $a>0$ and opens down if $a<0$.

- The parabola is wider than the graph of $y = x^2$ if $|a| < 1$ and narrower than the graph of $y = x^2$ if $|a| > 1$.

- The x-coordinate of the vertex is $-b/(2a)$.

- The axis of symmetry is the vertical line $x = -b/(2a)$.



EXAMPLE

Graph $y = 2x^2 - 8x + 6$

Solution: The coefficients for this function are
$a = 2$, $b = -8$, $c = 6$.

Since $a>0$, the parabola opens up.

The x-coordinate is: $x = -b/2a$, $x = -(-8)/(2(2))$

$x = 2$

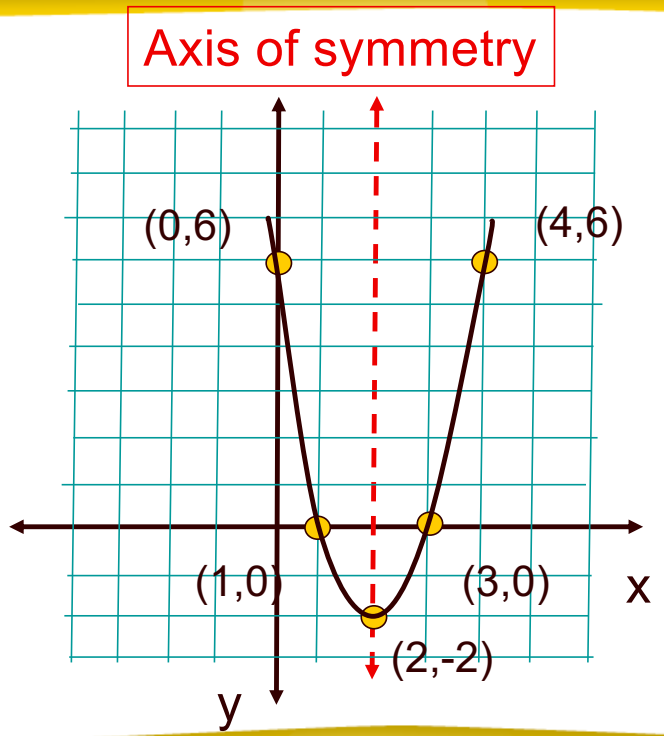
The y-coordinate is: $y = 2(2)^2-8(2)+6$

$y = -2$

Hence, the vertex is $(2,-2)$.

EXAMPLE (contd.)

- Draw the vertex (2,-2) on graph.
- Draw the axis of symmetry x=-b/(2a).
- Draw two points on one side of the axis of symmetry such as (1,0) and (0,6). How were these points chosen?
- Use symmetry to plot two more points such as (3,0), (4,6).
- Draw parabola through the plotted points.



VERTEX FORM OF QUADRATIC EQUATION

$$y = a(x - h)^2 + k$$

- The vertex is (h,k).
- The axis of symmetry is x = h.

GRAPHING A QUADRATIC FUNCTION IN VERTEX FORM

- Example $y = -1/2(x + 3)^2 + 4$

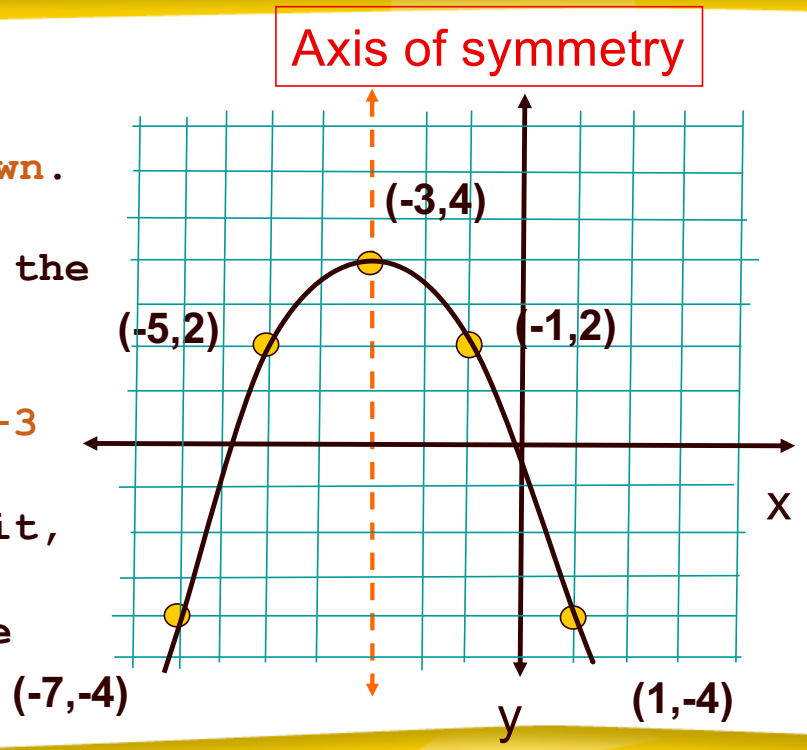
where a = -1/2, h = -3, k = 4.

Since a<0, the parabola opens down.

- To graph a function, first plot the vertex (h,k) = (-3,4).

- Draw the axis of symmetry x = -3

- Plot two points on one side of it, such as (-1,2) and (1,-4).
- Use the symmetry to complete the graph.



INTERCEPT FORM OF QUADRATIC EQUATION

$$y = a(x - p)(x - q)$$

- The x intercepts are p and q.

- The axis of symmetry is halfway between (p,0) and (q,0).

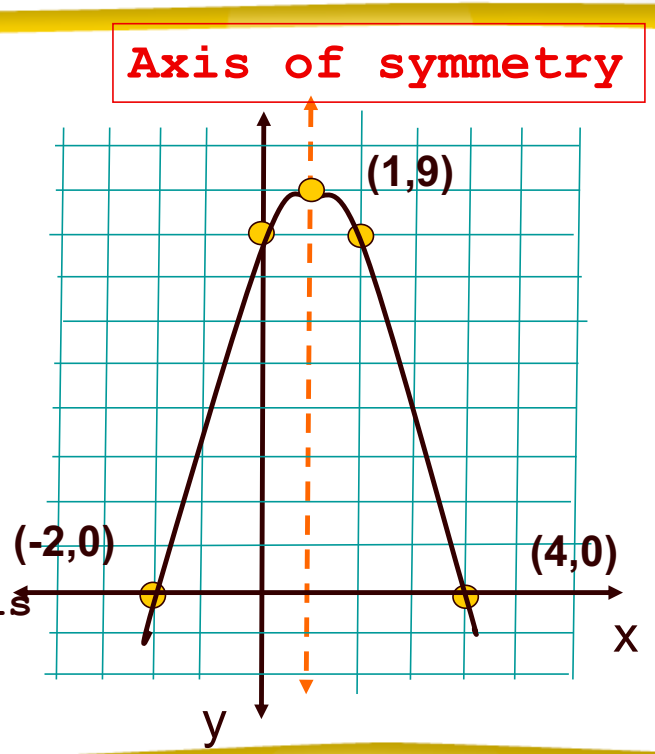
GRAPHING A QUADRATIC FUNCTION IN INTERCEPT FORM

- Example $y = -(x + 2)(x - 4)$. where $a = -1$, $p = -2$, $q = 4$. Since $a<0$ the parabola opens down.

- To graph a function, the x-intercepts occur at $(-2,0)$ and $(4,0)$.

- Draw the axis of symmetry that lies halfway between these points at $x = 1$.

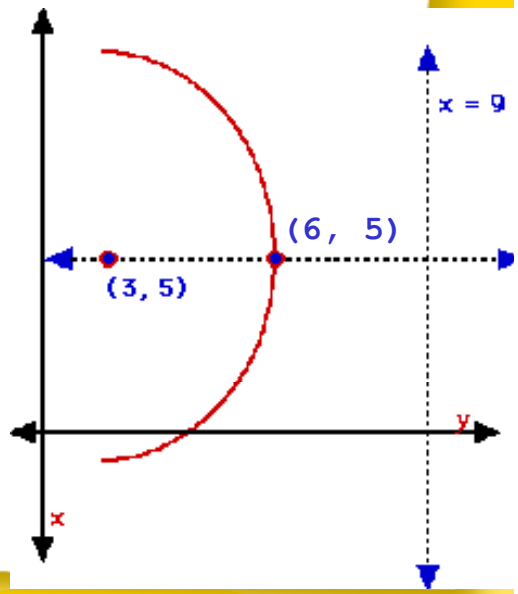
- So, the x - coordinate of the vertex is $x = 1$ and the y - coordinate of the vertex is: $y = -(1 + 2)(1 - 4) = 9$.



Finding the Equations of Parabolas

Write the equation of the parabola with a focus at (3, 5) and the directrix at $x = 9$, in standard form and general form

The distance from the focus to the directrix is 6 units, therefore, $2p = -6$, $p = -3$. Thus, the vertex is (6, 5).



The axis of symmetry is parallel to the x-axis:

$(y - k)^2 = 4p(x - h)$

$(y - 5)^2 = 4(-3)(x - 6)$ $h = 6$ and $k = 5$

$(y - 5)^2 = -12(x - 6)$ Standard form

Finding the Equations of Parabolas

Find the equation of the parabola that has a minimum at (-2, 6) and passes through the point (2, 8).

The axis of symmetry is parallel to the y-axis.

The vertex is (-2, 6), therefore, h = -2 and k = 6.

Substitute into the standard form of the equation and solve for p:

$(x - h)^2 = 4p(y - k)$ x = 2 and y = 8

$(2 - (-2))^2 = 4p(8 - 6)$

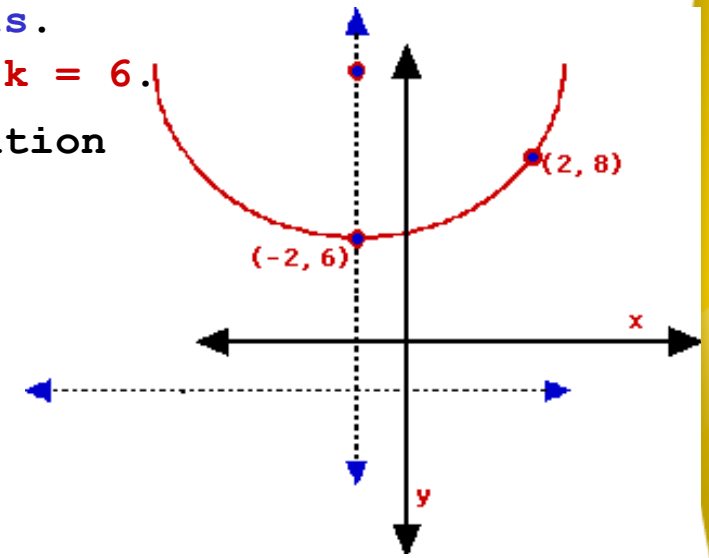
$16 = 8p$

$2 = p$

$(x - h)^2 = 4p(y - k)$

$(x - (-2))^2 = 4(2)(y - 6)$

$(x + 2)^2 = 8(y - 6)$ Standard form



Analyzing a Parabola

Find the coordinates of the vertex and focus, the equation of the directrix, the axis of symmetry, and the direction of opening of $2x^2 + 4x - 2y + 6 = 0$.

$2x^2 + 4x - 2y + 6 = 0$

$2(x^2 + 2x + \underline{\quad 1 \quad}) = 2y - 6 + \underline{\quad 2(1) \quad}$

$2(x + 1)^2 = 2(y - 2)$

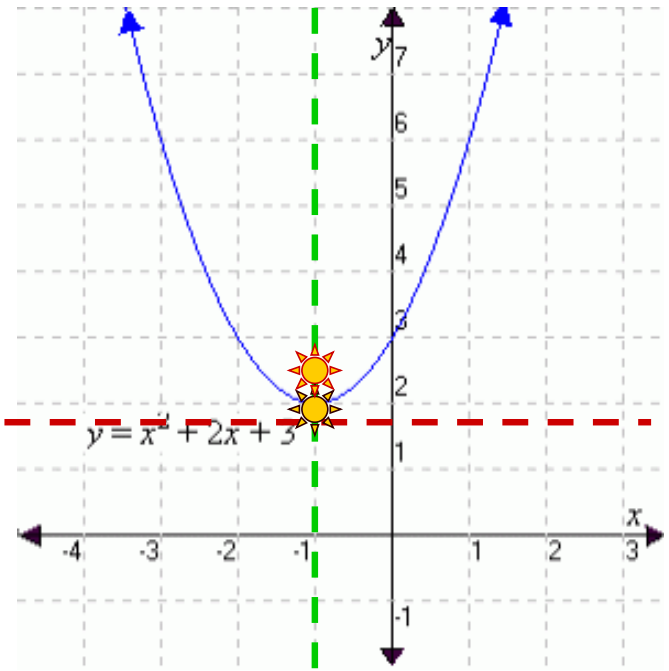
$(x + 1)^2 = (y - 2)$

$4p = 1$

$p = \frac{1}{4}$

Analyzing a Parabola

The parabola opens to upward.
The vertex is (-1, 2).
The focus is (-1, 2 ¼).
The Equation of directrix is y = 1¾
The axis of symmetry is x = -1 .



Graphing a Parabola

$y^2 - 10x + 4y - 16 = 0$

$y^2 + 4y + \underline{\quad 4 \quad} = 10x + 16 + \underline{\quad 4 \quad}$

$(y + 2)^2 = 10x + 20$

$(y + 2)^2 = 10(x + 2)$

Horizontally oriented (right)

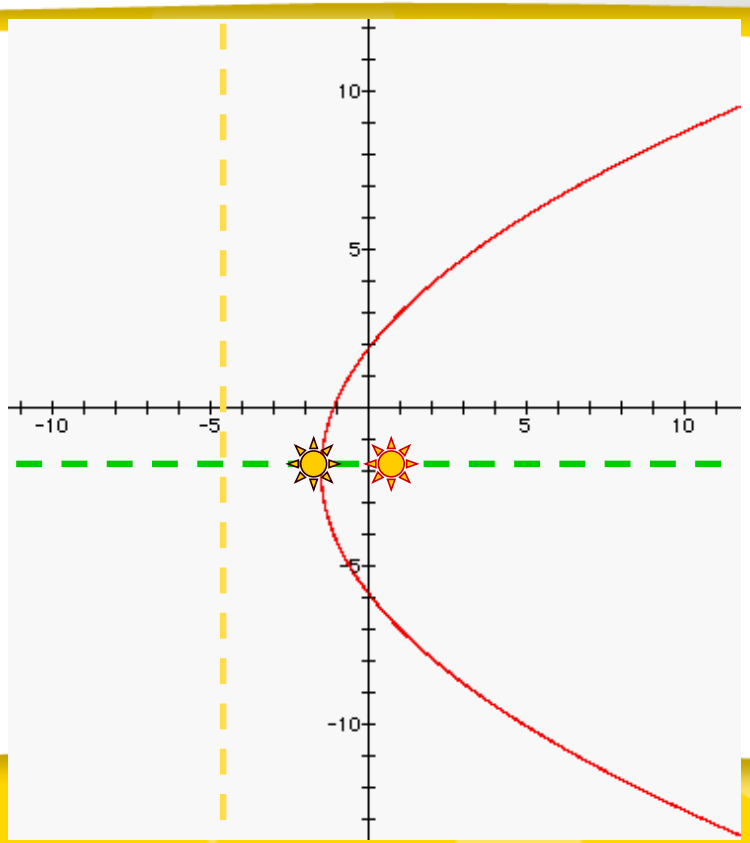
Vertex @ (-2, -2)

Line of Symmetry y = -2

P = 2.5

focus @ (0.5, -3)

Directrix X = - 4.5



WRITING THE QUADRATIC EQUATION IN STANDARD FORM

(1). $y = -(x + 4)(x - 9)$

$= -(x^2 - 9x + 4x - 36)$

$= -(x^2 - 5x - 36)$

$= -x^2 + 5x + 36$

(2). $y = 3(x - 1)^2 + 8$

$= 3(x - 1)(x - 1) + 8$

$= 3(x^2 - x - x + 1) + 8$

$= 3(x^2 - 2x + 1) + 8$

$= 3x^2 - 6x + 3 + 8$

$= 3x^2 - 6x + 11$

You Try!!

Write the following equation in parabolic form. State the vertex, axis of symmetry and direction of opening.

$$x = y^2 - 10y - 7$$

Parabolic form: $x = (y-5)^2 - 32$

Vertex: (-32, 5)

Axis of symmetry: $y = 5$

Direction of Opening: right

Equations of a Parabola

$x = ay^2 + by + c$ Horizontal Hyperbola	$y = ax^2 + bx + c$ Vertical Hyperbola
★ vertex: $y = \frac{-b}{2a}$	★ Vertex: $x = \frac{-b}{2a}$
★ If $a > 0$, opens right If $a < 0$, opens left	★ If $a > 0$, opens up If $a < 0$, opens down
★ The directrix is vertical	★ The directrix is horizontal

$$a = \frac{1}{4p}$$

Remember: $|p|$ is the distance from the vertex to the focus

the directrix is the same distance from the vertex as the focus is

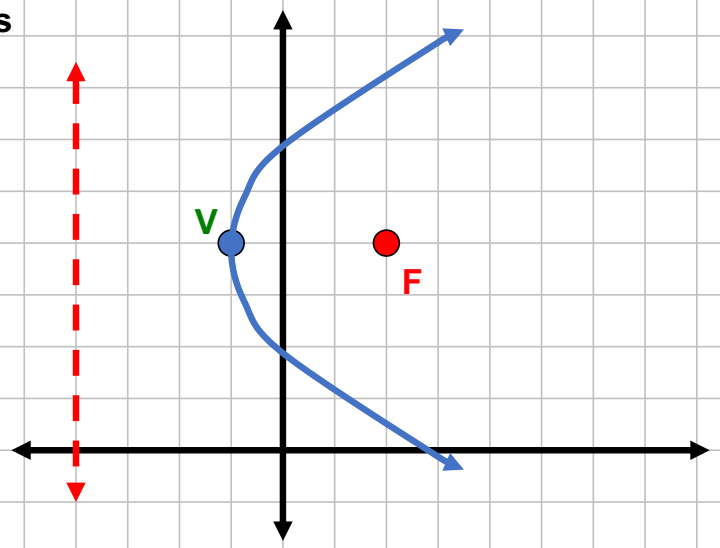
Standard Form Equation of a Parabola

$(y - k)^2 = 4p(x - h)$ **Horizontal Parabola**	$(x - h)^2 = 4p(y - k)$ **Vertical Parabola**
★ Vertex: (h, k)	★ Vertex: (h, k)
★ If $4p > 0$, opens right If $4p < 0$, opens left	★ If $4p > 0$, opens up If $4p < 0$, opens down
★ The directrix is vertical the vertex is midway between the focus and directrix	★ The directrix is horizontal and the vertex is midway between the focus and directrix

Remember: $|p|$ **is the distance from the vertex to the focus**

Find the standard form of the equation of the parabola given: the focus is (2, 4) and the directrix is x = - 4

The vertex is midway between the focus and directrix, so the vertex is (-1, 4)



The directrix is vertical so the parabola must be horizontal and since the focus is always inside the parabola, it must open to the right

Equation: $(y - k)^2 = 4p(x - h)$

$|p| = 3$

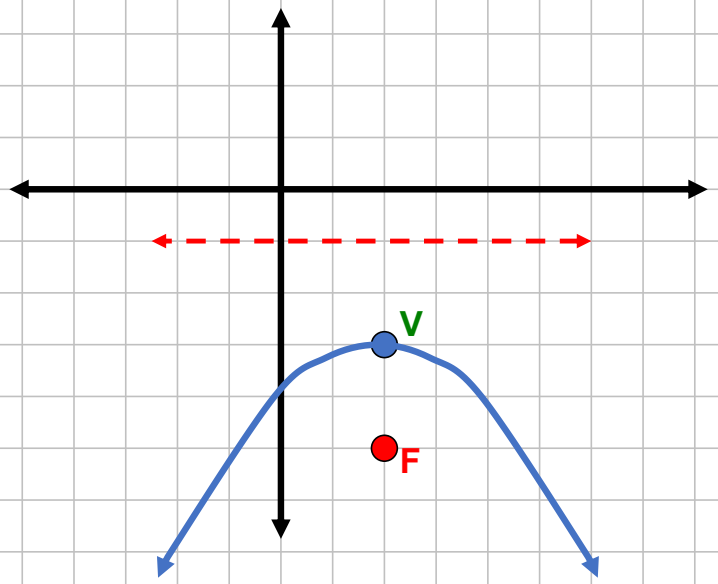
Equation: **$(y - 4)^2 = 12(x + 1)$**

Find the standard form of the equation of the parabola given: the vertex is (2, -3) and focus is (2, -5)

Because of the location of the vertex and focus this must be a vertical parabola that opens down

Equation: $(x - h)^2 = 4p(y - k)$

$|p| = 2$



Equation: $(x - 2)^2 = -8(y + 3)$

The vertex is midway between the focus and directrix, so the directrix for this parabola is y = -1

Graphing a Parabola

$(y + 3)^2 = 4(x + 1)$

Find the vertex, focus and directrix. Then graph the parabola

Vertex: (-1, -3)

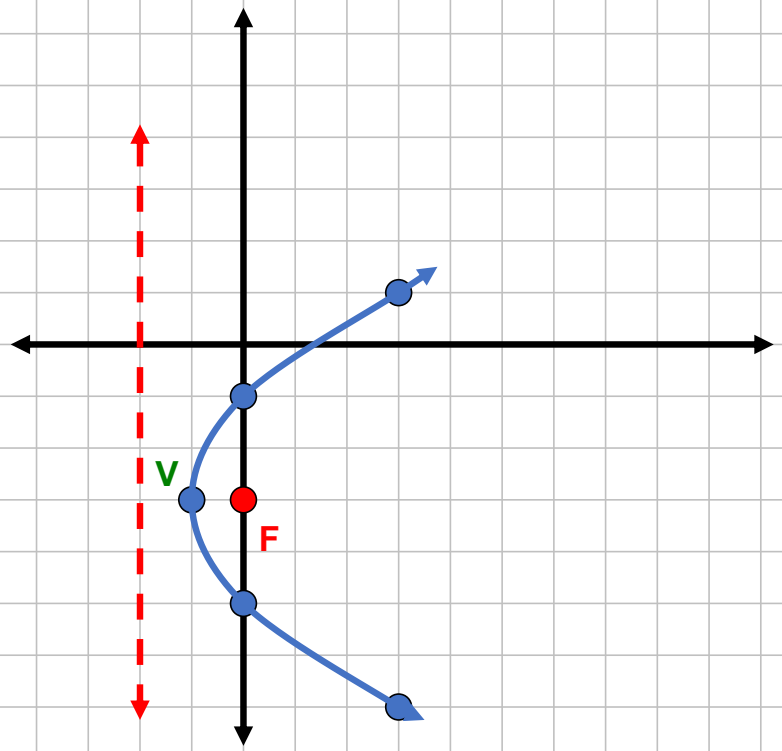
The parabola is horizontal and opens to the right

$4p = 4$

$p = 1$

Focus: (0, -3)

Directrix: x = -2



$x = \frac{1}{4}(y + 3)^2 - 1$

x	0	0	3	3
y	-1	-5	1	-7

Converting an Equation

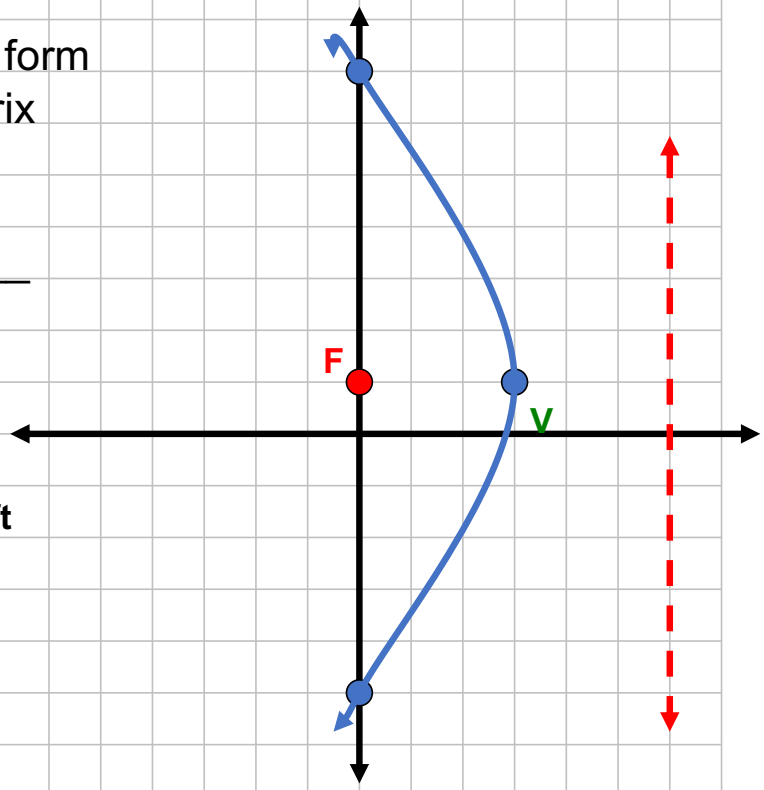
Convert the equation to standard form
Find the vertex, focus, and directrix

$y^2 - 2y + 12x - 35 = 0$

$y^2 - 2y + \underline{1} = -12x + 35 + \underline{1}$

$(y - 1)^2 = -12x + 36$

$(y - 1)^2 = -12(x - 3)$



The parabola is horizontal and opens left

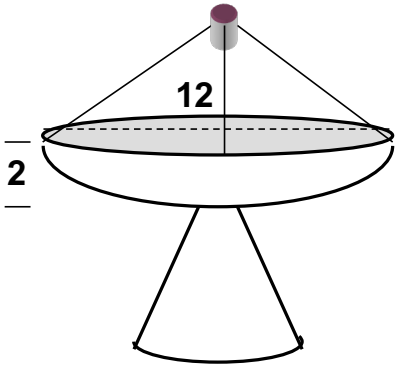
Vertex: (3, 1)

4p = -12
p = -3

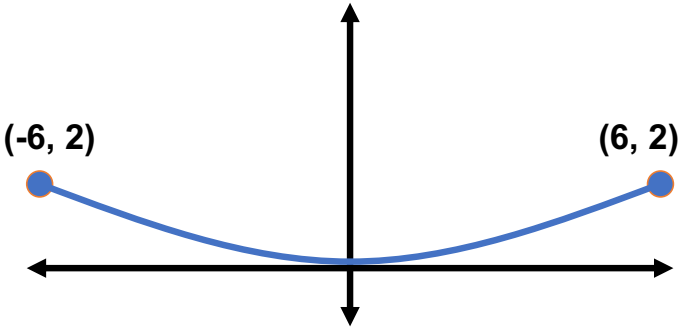
Focus: (0, 1)

Directrix: x = 6

Applications



A satellite dish is in the shape of a parabolic surface. The dish is 12 ft in diameter and 2 ft deep. How far from the base should the receiver be placed?



Consider a parabola cross-section of the dish and create a coordinate system where the origin is at the base of the dish.

Since the parabola is vertical and has its vertex at (0, 0) its equation must be of the form:

$$x^2 = 4py$$

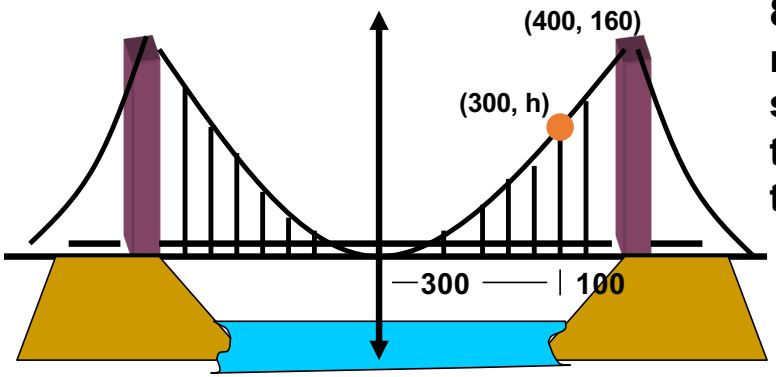
At (6, 2), $36 = 4p(2)$

so $p = 4.5$

thus the focus is at the point (0, 4.5)

The receiver should be placed 4.5 feet above the base of the dish.

Application



The towers of a suspension bridge are 800 ft apart and rise 160 ft above the road. The cable between them has the shape of a parabola, and the cable just touches the road midway between the towers.

What is the height of the cable 100 ft from a tower?

Since the parabola is vertical and has its vertex at (0, 0) its equation must be of the form:

$$x^2 = 4py$$

At (400, 160), $160{,}000 = 4p(160)$

$1000 = 4p$

$p = 250$

thus the equation is $x^2 = 1000y$

At (300, h), $90{,}000 = 1000h$

$h = 90$

The cable would be 90 ft long at a point 100 ft from a tower.

THE END.

Graph Quadratic Functions

Example Consider the quadratic function $y = x^2 + 8x - 12$.

a. Determine whether the parabola opens upward or downward.
b. Find the y-intercept.
c. Find the vertex.
d. Find the equation of the axis of symmetry.
e. Find the x-intercepts, if any.
f. Draw the graph.

Graph Quadratic Functions

a. Since a is -1, which is less than 0, the parabola opens downward.

b. To find the y-intercept, set x = 0 and solve for y.

$$y = -(0)^2 + 8(0) - 12 = -12$$

The y-intercept is (0, 12)

Graph Quadratic Functions

c. First, find the x-coordinate, then find the y-coordinate of the vertex. From the function, a = -1, b = 8, and c = -12.

$$x = -\frac{b}{2a} = \frac{8}{2(-1)} = 4$$

Since the x-coordinate of the vertex is not a fraction, we will substitute x = 4 into the original function to determine the y-coordinate of the vertex.

$$y = -x^2 + 8x - 12$$

$$y = -(4)^2 + 8(4) - 12$$

$$= -16 + 32 - 12 = 4$$

The vertex is (4, 4).

Graph Quadratic Functions

d. Since the axis of symmetry is a vertical line through the vertex, the equation is found using the same formula used to find the x-coordinate of the vertex (see part c).

Thus, the equation of the axis of symmetry is $x = 4$.

Graph Quadratic Functions

e. To find the x-intercepts, set y = 0.

$$x^2 - 8x + 12 = 0$$

$$(x-6)(x-2) = 0$$

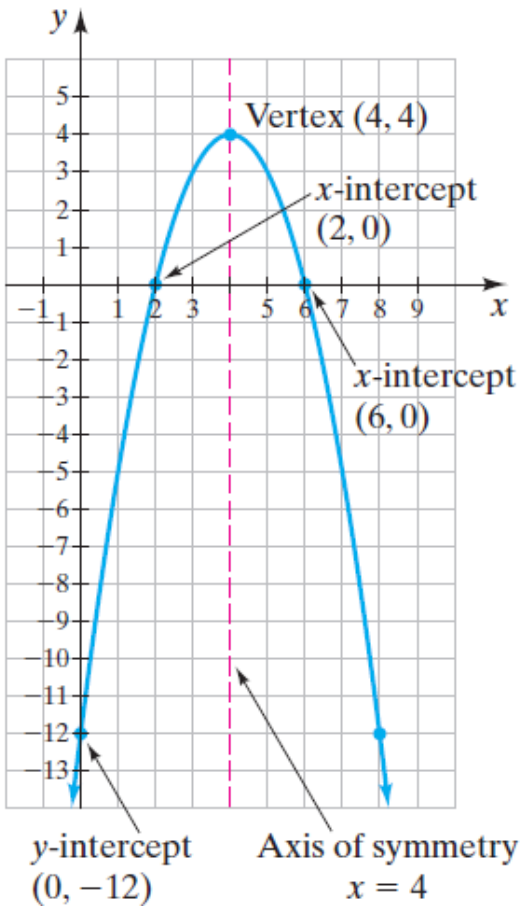
$$x - 6 = 0 \quad \text{or} \quad x - 2 = 0$$

$$x = 6 \qquad\qquad x = 2$$

Thus, the x-intercepts are (2, 0) and (6, 0). These values could also be found by the quadratic formula (or by completing the square).

Graph Quadratic Functions

f. Draw the graph.

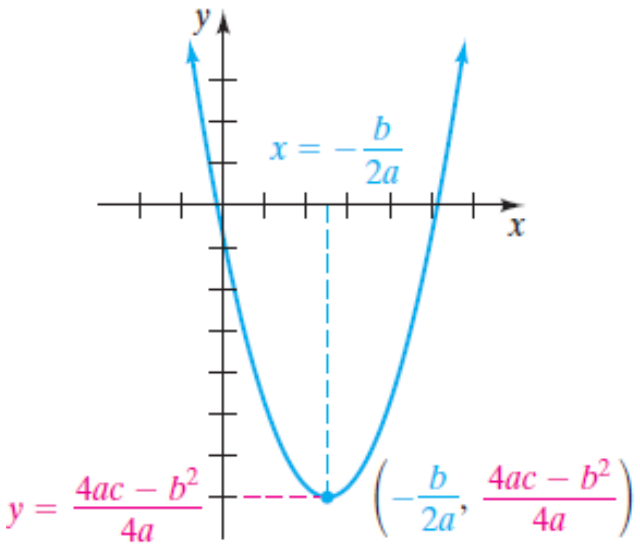


Solve Maximum and Minimum Problems

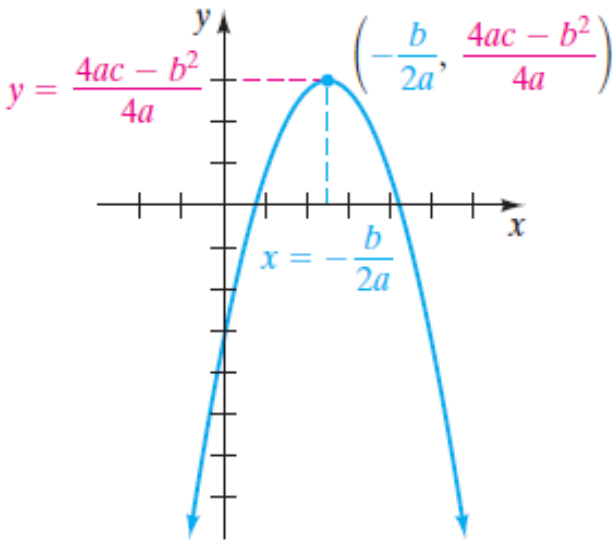
A parabola that opens upward has a minimum value at its vertex, and a parabola that opens downward has a maximum value at its vertex.

$y = ax^2 + bx + c$

$a > 0$, minimum value

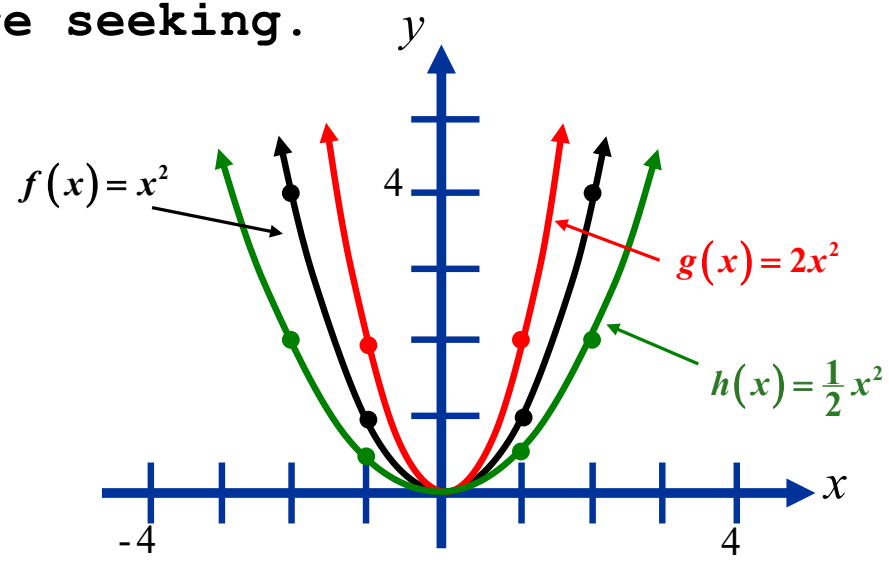


$a < 0$, maximum value



Understand Translations of Parabolas

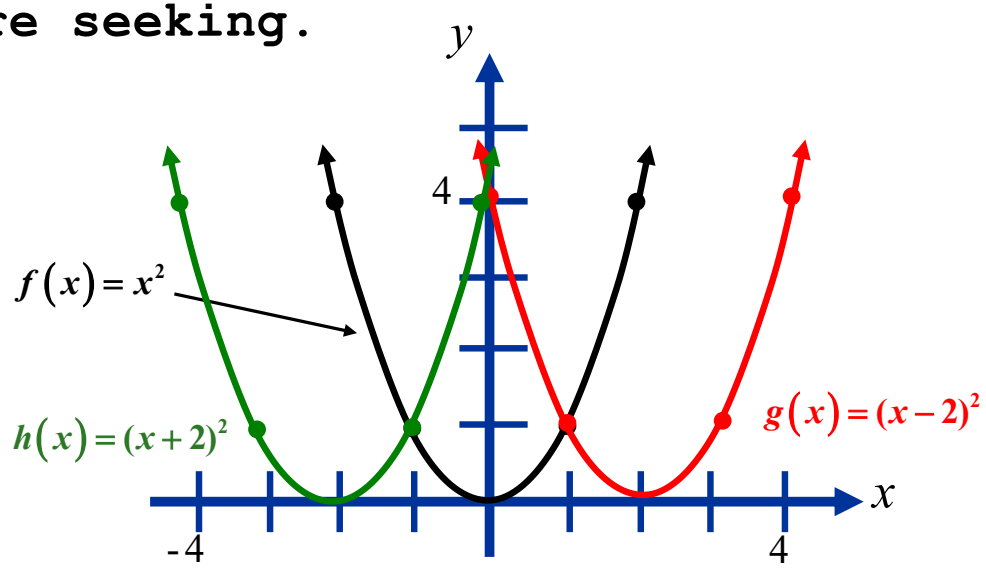
Start with the basic graph of $f(x) = ax^2$ and translate, or shift, the position of the graph to obtain the graph of the function you are seeking.



Notice that the value of a in the graph $f(x) = ax^2$ determines the width of the parabola. As |a| gets larger, the parabola gets narrower, and as |a| gets smaller, the parabola gets wider.

Understand Translations of Parabolas

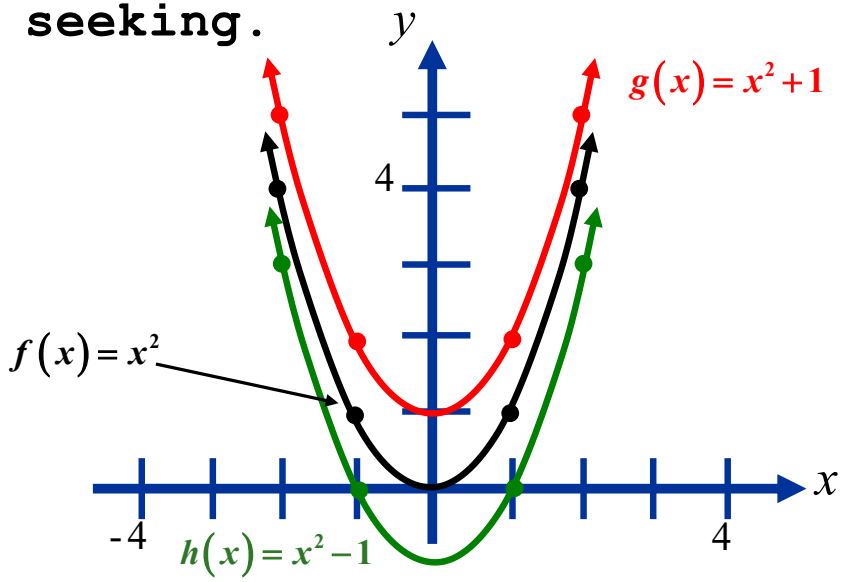
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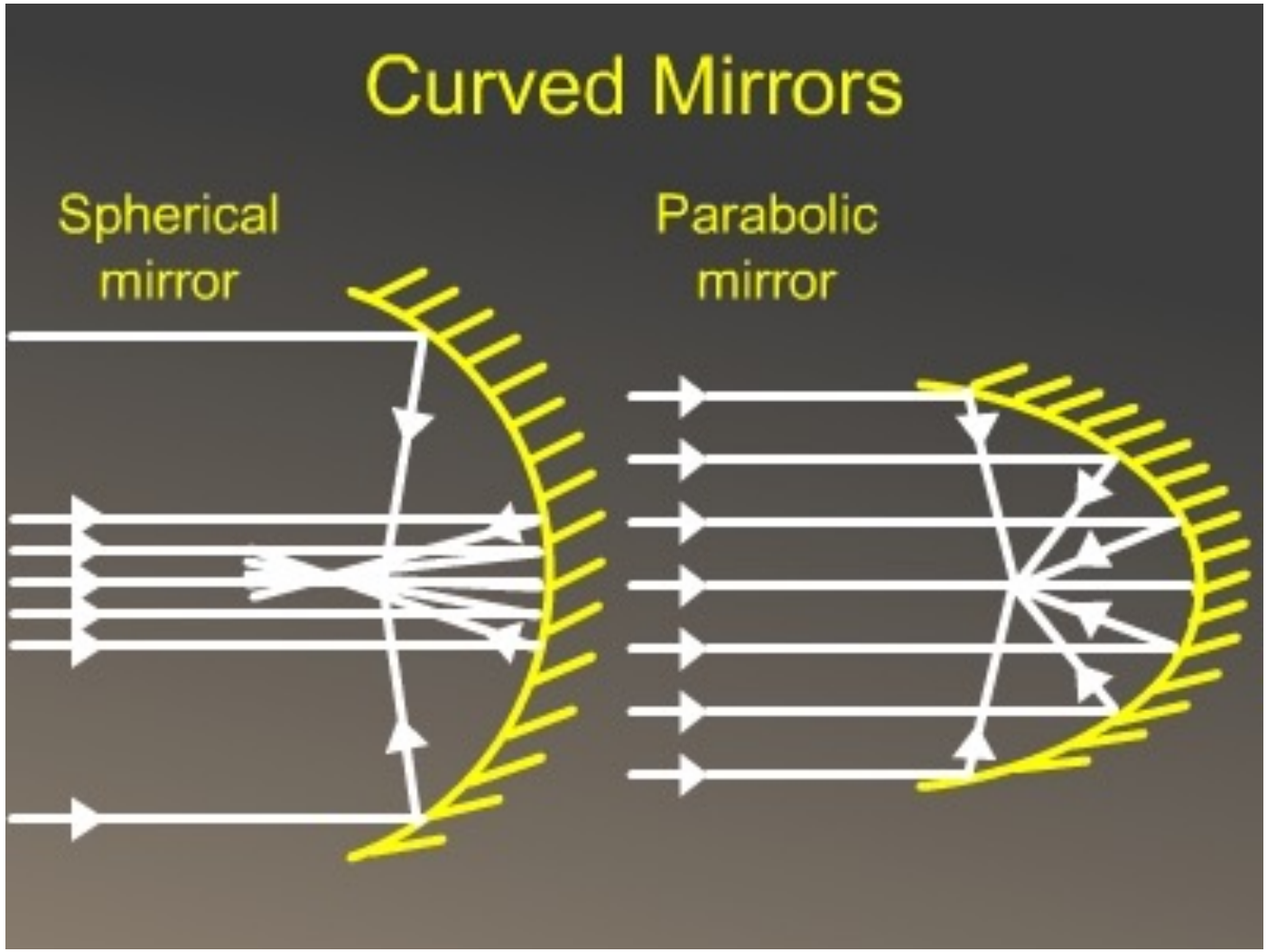
If h is a positive real number, the graph of $g(x) = a(x - h)^2$ will be shifted h units to the right of the graph $g(x) = ax^2$. If h is a negative real number, the graph of $g(x) = a(x - h)^2$ will be shifted |h| units to the left.

Understand Translations of Parabolas

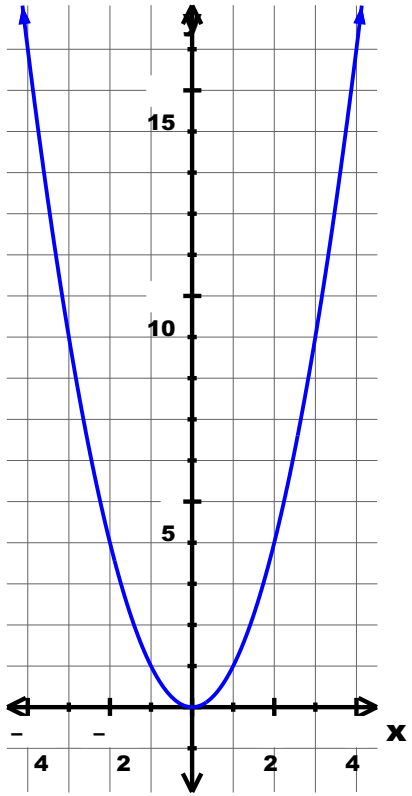
Start with the basic graph of $f(x) = ax^2$ and translate, or shift, the position of the graph to obtain the graph of the function you are seeking.



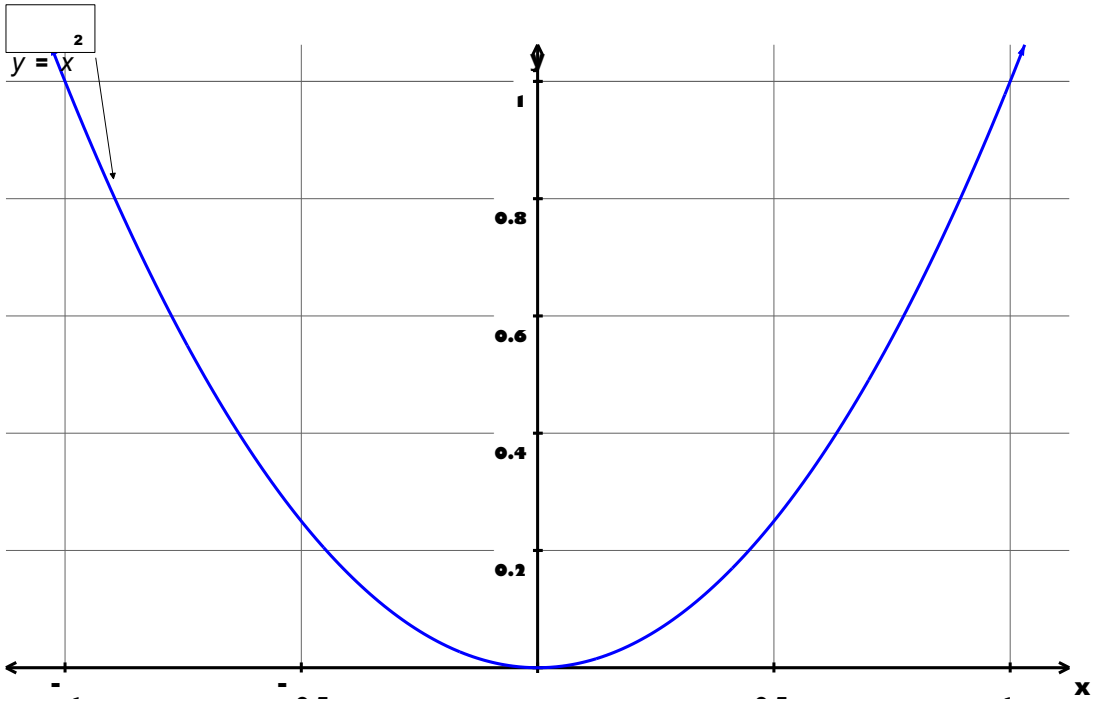
In general, the graph of $g(x) = ax^2 + k$ is the graph of $f(x) = ax^2$ shifted k units up if k is a positive real number and $|k|$ units down if k is a negative real number.


Curved Mirrors
Spherical mirror
Parabolic mirror

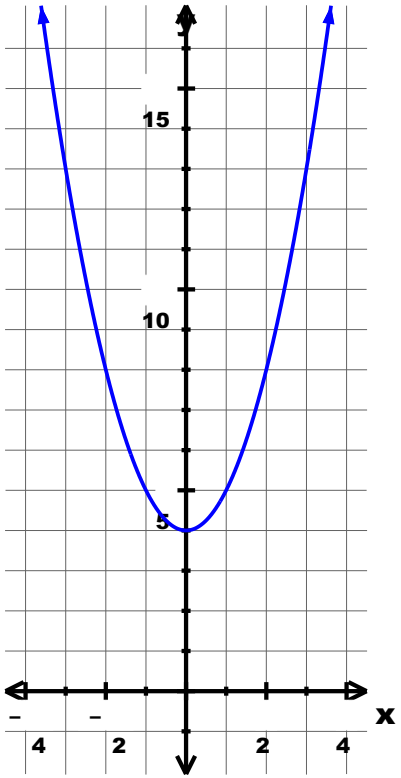
$y=x^2$



$y=x^2$ close up

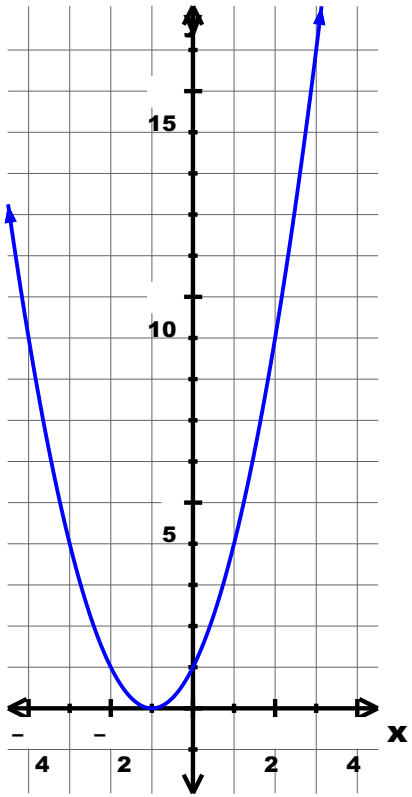


Vertical Translation



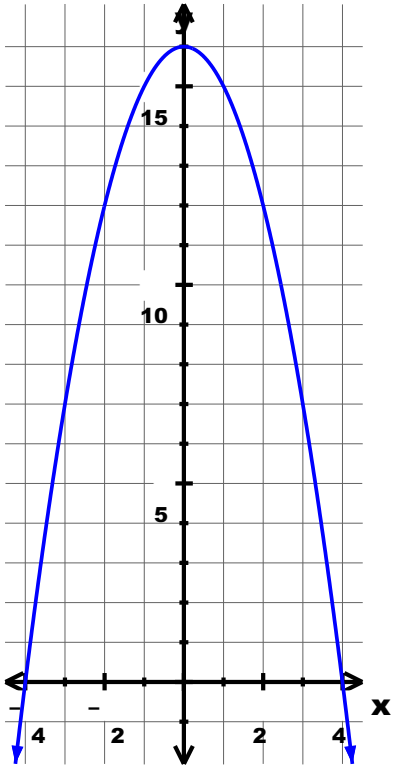
$$y = x^2 + 4$$

Horizontal Translation



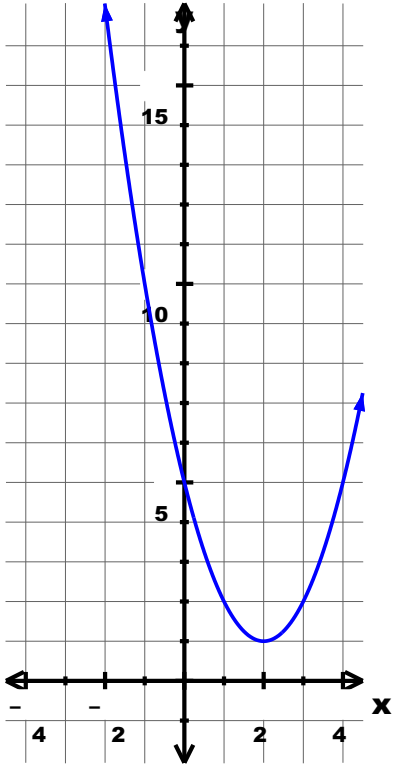
$$y = (x+1)^2$$

Reflection in a horizontal axis



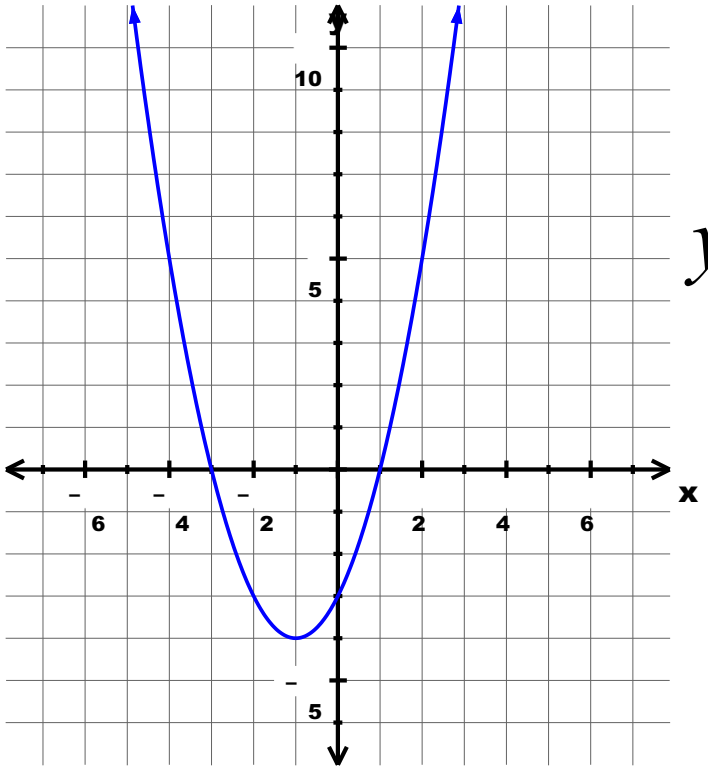
$$y = 16 - x^2$$

Combinations of translations



$$y = (x-2)^2 + 1$$

Factorised form



$$y = (x-1)(x+3)$$